

Artificial Intelligence

BSc Computer Science - Durham University

Matthew Emerson

October 2026

Contents

1	Module Introduction	1
1.1	Assessment	1
1.2	Timetable	1
2	Machine Learning	1
2.1	Introduction to Machine Learning	1
2.2	Machine Learning Workflow and Coding Environment	1
2.3	Data Understanding	1
2.4	Data Preparation Workflow	1
3	Artificial Intelligence Search	1
3.1	Coding Real-World Problems for Search Tree Expansion	1
3.2	Uninformed Search: BFS, DFS, and Iterative Deepening	2
3.3	General Heuristic Search: Greedy Best-First Search and A* Search	2
3.4	A* Search and Deriving Iterative Deepening A* Search	2
3.5	Heuristic Functions and Their Development	2
3.6	Local Search: Hill-Climbing, Simulated Annealing, and Genetic Algorithms	2
3.7	Swarm Behaviour, How Ants Forage, and a Simple S-ACO algorithm	2
3.8	Applying Different Ant Systems to the TSP	2
3.9	General Versions of Particle Swarm Optimisation and Differential Evolution	2
3.10	Applying Particle Swarm Optimization to the TSP	2
4	Ethics and Bias in AI	2
4.1	Introduction to AI Ethics and Related Concepts	2
4.2	Societal Impacts of AI	2
4.3	Defining Bias and Fairness in ML	2
4.4	How to Build Better Systems	2
4.5	Value Sensitive Design	2
4.6	Evaluations and Audits	2
4.7	Bias Mitigation Strategies	2
4.8	Transparency and Accountability for Trustworthy Systems	2
4.9	Explainable AI	2
4.10	Future Trends and Ethical Frontiers	2
I	Appendix	3
5	Bibliography	3
5.1	Module Professors	3
5.2	References	3
5.3	List of Figures	3
6	Reading List	3
6.1	Machine Learning	3
6.2	Ethics and Bias in AI	3

1 Module Introduction

Artificial Intelligence has the module code COMP2261, and is led by Dr Yang Long.

1.1 Assessment

This module is assessed through three pieces of summative coursework, these can be seen in Table 1. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Machine Learning	10/10/2026	16/12/2026	22/01/2026	Dr Yang Long	50%
AI Search	12/01/2027	19/02/2027	19/03/2026	Prof. Iain Stewart	25%
Bias in AI	06/02/2027	26/03/2027	07/05/2027	Dr. Shauna Concannon	25%

Table 1: Assessments for Artificial Intelligence.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	14:00 - 16:00	Computer Class	TLC025	n/a
Tuesday	16:00 - 17:00	Lecture	W103	Dr Yang Long
Friday	15:00 - 16:00	Lecture	W103	Dr Yang Long

Table 2: Michaelmas Term Timetable for Artificial Intelligence.

2 Machine Learning

This section is taught by Dr Yang Long. The recommended reading for this topic is:

- Mathematics for Machine Learning (Deisenroth, Faisal and Ong, 2020).
- Mastering Machine Learning with scikit-learn (Hackeling, 2017).

2.1 Introduction to Machine Learning

2.2 Machine Learning Workflow and Coding Environment

2.3 Data Understanding

2.4 Data Preparation Workflow

3 Artificial Intelligence Search

This section is taught by Professor Iain Stewart. There is no recommended reading for this topic however there may be an additional video to watch after lectures to enable more content to be covered.

Watch the FAQs video.

3.1 Coding Real-World Problems for Search Tree Expansion

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.2 Uninformed Search: BFS, DFS, and Iterative Deepening

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.3 General Heuristic Search: Greedy Best-First Search and A* Search

3.4 A* Search and Deriving Iterative Deepening A* Search

3.5 Heuristic Functions and Their Development

3.6 Local Search: Hill-Climbing, Simulated Annealing, and Genetic Algorithms

3.7 Swarm Behaviour, How Ants Forage, and a Simple S-ACO algorithm

3.8 Applying Different Ant Systems to the TSP

3.9 General Versions of Particle Swarm Optimisation and Differential Evolution

3.10 Applying Particle Swarm Optimization to the TSP

4 Ethics and Bias in AI

This section is taught by Dr Shauna Concannon. The recommended reading is handled on a weekly basis for this topic.

4.1 Introduction to AI Ethics and Related Concepts

4.2 Societal Impacts of AI

4.3 Defining Bias and Fairness in ML

4.4 How to Build Better Systems

4.5 Value Sensitive Design

4.6 Evaluations and Audits

4.7 Bias Mitigation Strategies

4.8 Transparency and Accountability for Trustworthy Systems

4.9 Explainable AI

4.10 Future Trends and Ethical Frontiers

Part I

Appendix

5 Bibliography

5.1 Module Professors

Module Lead: Doctor Yang Long

- Focus: Scalable AI solutions for large-scale healthcare applications.
- Email: yang.long@durham.ac.uk
- Office Hours: Monday 15:30 - 16:30 in Term 1.

Lecturer: Professor Iain Stewart

- Focus: Computational complexity, graph/group theory, and theoretical aspects artificial intelligence.
- Email: i.a.stewart@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00 in Term 2.

Lecturer: Doctor Shauna Concannon

- Focus: Computational linguistics, social interaction, and human-computer interaction.
- Email: shauna.j.concannon@durham.ac.uk
- Office Hours: Thursday 14:30 - 15:30 in Term 2.

5.2 References

5.3 List of Figures

6 Reading List

6.1 Machine Learning

Deisenroth, M.P., Faisal, A.A. and Ong, C.S. (2020). Mathematics for Machine Learning. [online] Cambridge University Press. Available at: <https://mml-book.com/> [Accessed 3 Oct. 2026].

Hackeling, G. (2017). Mastering Machine Learning with scikit-learn. 2nd edn. Packt Publishing.

6.2 Ethics and Bias in AI

Various, will be added over time.